

A 0.26 μVrms instrumentation amplifier based on discrete-time DC servo loop with successive-approximation-compensation technique and Chopper DAC Array

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ABSTRACT

A low-noise instrumentation amplifier for bio-signal recording is proposed in this paper. One of the significant disturbances, electrode DC offset, is suppressed using the proposed successive-approximation-compensation (SAC) circuit based on a discrete-time DC servo loop with Chopper DAC Array. Firstly, the DC offset at the output is quantified using the SAR quantizer. Next, the digital code produced by the SAR quantizer is used to control the chopper DAC array. Finally, a suitable compensation voltage is generated and used directly to correct the electrode DC offset at the input of the chopper amplifier. By using the proposed compensation technique, the first-stage chopper amplifier's quiescent operating state is unaffected. When a 50-mV DC offset voltage exists at the differential input, the SAC circuit can compensate the DC offset voltage close to zero in a relatively short time. At the same time, there are fewer capacitors, which significantly decrease the area consumption. The proposed instrumentation amplifier is implemented in a standard 0.18 μm CMOS process. The experiment results show that over a bandwidth from 0.1 Hz to 100 Hz, an input referred noise of 0.26 μVrms at the TT corner has been attained. The total power consumption is about 13.72 μW at the supply voltage of 1.8 V. The gain of the instrumentation amplifier is about 39.15 dB and its Noise Efficiency Factor (NEF) is 2.76.

1. Introduction

With the rapid development of integrated circuit technology, wearable medical electronic devices play an increasingly important role in people's daily life. Bio-signal monitoring equipment can monitor human health in real time by continuously acquiring human bio-signals (e.g. EEG, ECG, EMG) [1]. However, since the bio-signals of the human body have a series of particularities (tiny amplitude, extremely low frequency, and extremely narrow bandwidth) [2], we face great challenges in improving signal processing accuracy and reducing chip area.

One of the disturbances that need to be overcome is the electrode DC offset [3,4]. $V_{HC1,2}$ represents the half-cell potential on the electrode in Fig. 1. $V_{HC1,2}$ of two different electrodes are not completely same, because the contact between the skin and the electrode is uncertain. Even V_{HC} of the same electrode will change slowly over time with the state of the human body [5]. Therefore, a differential DC offset voltage will be generated on the input node of the analog front-end acquisition circuit, which may saturate the output of the chopper instrumentation amplifier [6]. Thereby, this will affect the

normal operation of this overall circuit. In general, wet electrodes (e.g. Ag/AgCl) have an electrode offset voltage of about 50 mV, while dry electrodes (metal discs, flexible films, etc.) can reach 300 mV [7]. Therefore, to improve the quality of the signal collected by the analog front-end circuit, the DC offset voltage should be eliminated as much as possible.

To eliminate the electrode DC offset, some off-chip elements are employed at the input stages [8,9] which will bring a great inconvenience. In addition, some AC-coupled techniques are used for this purpose [10,11]. However, the ultra-large resistance realized by MOS pseudo resistors will probably lead to large nonlinear. In previous work, a diode-capacitor network is also used to filter the DC polarization [12]. But it is difficult to control the stability of the circuit because of its open-loop design. The gate-controlled NMOS pseudo resistor is used in some papers [13,14] to obtain a tunable low-cutoff frequency, but the best value is not below 10 Hz. In recent years, the DC servo loop (DSL) is introduced to suppress the electrode DC offset [15], but the large capacitor area greatly increases the cost of tape-out.

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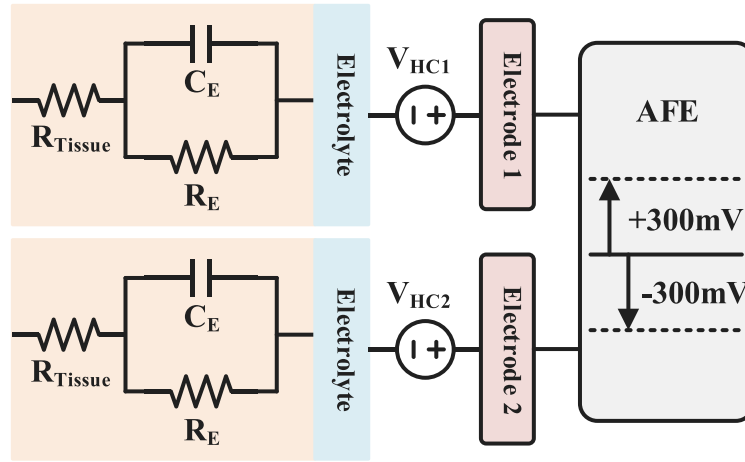


Fig. 1. Schematic diagram of the electrode DC offset.

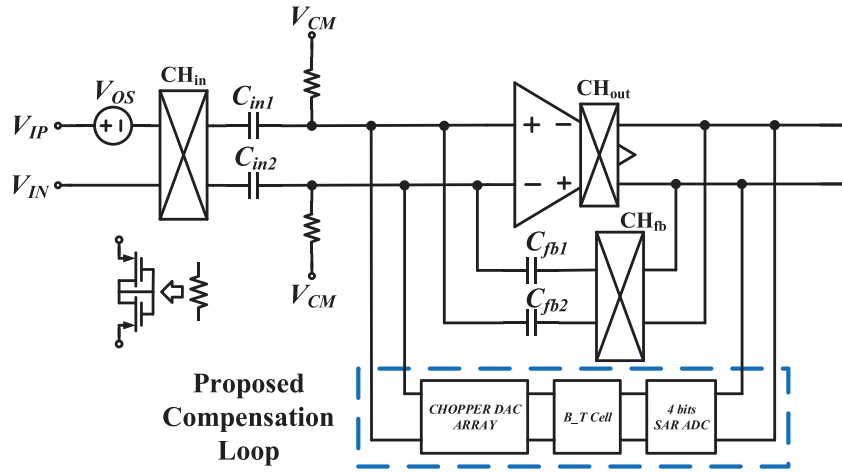


Fig. 2. Architecture diagram of the proposed SAC scheme.

In this paper, the successive-approximation-compensation (SAC) method is proposed to suppress the electrode DC offset at the input end. The main innovation of this new design is that the compensation voltage is generated by the chopper DAC array and used directly to compensate the electrode DC offset at the input of the chopper amplifier without affecting the quiescent operating point inside the main chopper amplifier. Secondly, compared with the traditional analog DSL compensation scheme, the circuit structure of this scheme is simpler, and the number of capacitors is less, so the occupied chip area is reduced, and its response speed is faster. The organization of this paper is as follows. In Section 2, the architectural overview of the SAC method is discussed. Section 3 analyzes the system-level compensation process of this scheme. Section 4 describes the key circuit modules in detail. Section 5 and Section 6 present the simulation results and conclusions, respectively.

2. Architectural overview of the proposed compensation scheme

When input signals pass through the input chopper switch, the electrode DC offset is removed from the baseband. The DC suppression structure by using the pseudo-resistance and the feedback capacitor cannot achieve the DC elimination effect. Therefore, the SAC scheme is adopted in this paper to eliminate the DC offset of the input electrode. The specific circuit implementation of the proposed SAC scheme is shown in Fig. 2, which consists of a capacitively coupled chopper instrumentation amplifier (CCIA), a 4-bit SAR quantizer, a binary-to-thermometer cell (B-T cell), and a chopper DAC array.

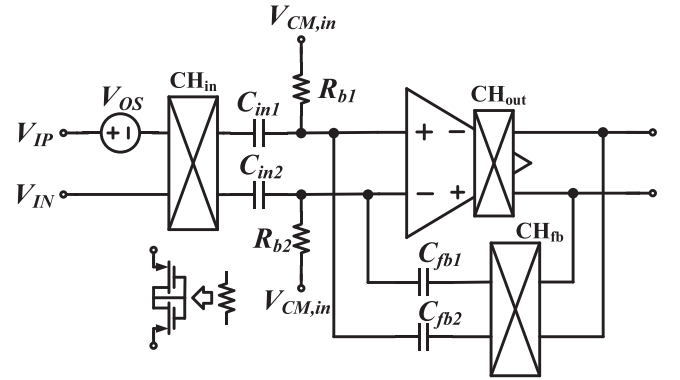


Fig. 3. Structure diagram of the capacitively coupled chopper instrumentation amplifier.

Firstly, the 4-bit SAR quantizer is used to quantize the output signal of the chopper amplifier and generate a 4-bit binary code, and then the 4-bit binary code is converted into a corresponding 15-bit thermometer code which is used to control the closure of the switches in the chopper DAC array. Finally, the chopper DAC array will generate a suitable compensation voltage that is fed back to the input of the chopper amplifier to eliminate the electrode DC offset voltage. This feedback loop eliminates the DC offset associated with input chopper non-ideal

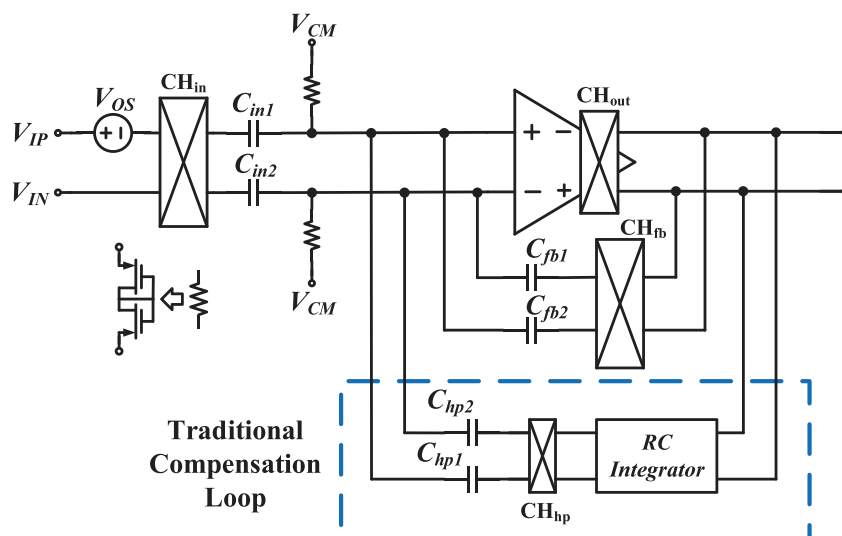


Fig. 4. Architecture diagram of the traditional analog DSL.

factors by the compensation voltage, thereby the offset voltage of the chopper amplifier output is reduced which avoids output saturation and malfunction.

3. Analysis of the system-level compensation process

The instrumentation amplifier is used as the input stage of the analog front-end circuit, and the performance of the overall circuit such as input impedance, common mode rejection ratio (CMRR), signal noise ratio (SNR), and power supply rejection ratio (PSRR) is determined by it. There are three traditional topologies of instrumentation amplifiers, which are the three-op-amp instrumentation amplifier structure, the current-mode structure, and the resistor-feedback structure. However, the closed-loop gain of these three topologies is determined by a resistor network loaded on the instrumentation amplifier output stage, so the circuits will consume a lot of power.

In this paper, a capacitively coupled chopper instrumentation amplifier (CCIA) is used, as shown in Fig. 3. The CCIA consists of an input chopper CH_{in} , an output chopper CH_{out} , a feedback chopper CH_{fb} , a capacitive feedback network, and an op-amp. The CCIA amplifies the input signal by a factor of C_{in}/C_{fb} to the output while modulating the offset of the op-amp input end and 1/f noise away from the base frequency. Finally, the up-modulated signal will be filtered out by the low-pass filter of the subsequent stage. R_{b1} and R_{b2} are two pseudo-resistors, whose resistance varies greatly with the process, leakage current, etc. Generally, the resistance value of the pseudo-resistor can change by an order of magnitude, so the pseudo-resistor bias is usually adjusted externally to make the pseudo-resistor resistance meet the design requirements and provide a suitable DC bias voltage $V_{CM,in}$ for the op-amp input end.

Compared with the three traditional instrumentation amplifier topologies, CCIA has many advantages. Firstly, CCIA has rail-to-rail sensing capability; Secondly, the accurate lithography of deep submicron technology results in good matching between C_{in} and C_{fb} , which in turn ensures high gain accuracy without the need for trimming. Finally, the noise of the CCIA mainly comes from the op-amp, so it is more power efficient than the three-op-amp structure and the current-mode structure.

In previous work, an analog DC servo loop (DSL) is usually used to suppress the DC offset of the input end. As shown in Fig. 4, the traditional analog DSL consists of an integrator, a chopper, and capacitors. Firstly, the integrator amplifies the DC signal at the output end of the CCIA, and then the chopper up-modulates the amplified DC signal. Finally, the up-modulated signal will be fed back to the input end

through capacitive coupling to eliminate the DC offset of the sampling electrode. The above integrator is generally a traditional RC structure. In the modern CMOS process, the matching degree and precision of the resistors are extremely poor. At the same time, due to the large capacitor area, this also greatly increases the cost of tape-out.

Therefore, the SAC method is proposed to eliminate the electrode DC offset at the input end. Fig. 5 illustrates the basic idea of the SAC scheme. The electrode DC offset V_{OS} at the input end will be amplified by the chopper instrumentation amplifier C_{in}/C_{fb} times to the output end, which may probably saturate the output of the chopper instrumentation amplifier, thereby affecting the normal operation of the overall circuit. Therefore, the circuit superposition theorem is applied here. A chopper DAC array is introduced at the input end to generate a DC voltage V_B opposite to the original electrode DC offset voltage. V_B will be amplified by the chopper instrumentation amplifier C_{in}/C_{fb} times to the output end. This will cancel the original DC offset out, and finally eliminate the influence of the electrode DC offset on the stability of the circuit. That is, such that:

$$V_{OS}C_{in1,2} = V_R C_{DSL} \quad (1)$$

where C_{DSL} is obtained by parallel connection of a certain number of C_p and C_n , and V_B is the difference between V_H and V_L . The above chopper DAC array can be simplified, which is composed of 15 identical modules. In each module, there will be a thermometer code (such as $TP\langle 1 \rangle$) to control the closing of switches S_1 and S_2 respectively, and then a series of different compensation voltages will be produced to eliminate the influence of electrode DC offset.

By analyzing the noise of this overall circuit, it can be seen that the noise of the instrumentation amplifier is the main noise source of this circuit and the noise of the proposed compensation loop accounts for only a small part of it. As shown in Fig. 5, the equivalent input noise of the overall circuit can be expressed as

$$\overline{V_{n,in}^2} \approx \overline{V_{n,OPA1}^2} \left(\frac{C_{IN} + C_{FB} + C_{HP}}{C_{IN}} \right)^2 + \overline{V_{n,DSL}^2} \left(\frac{C_{HP}}{C_{IN}} \right)^2 \quad (2)$$

where $\overline{V_{n,OPA1}^2}$ is the equivalent input noise of the input stage op-amp, C_{HP} is the sum of all unit capacitances in the chopper DAC array (i.e. the sum of all C_p and C_n), and $\overline{V_{n,DSL}^2}$ is the equivalent output noise of the proposed compensation loop.

4. Analysis of key circuit modules

In this section, the key circuit modules are analyzed in detail. Fig. 6 shows the generation process of the differential compensation

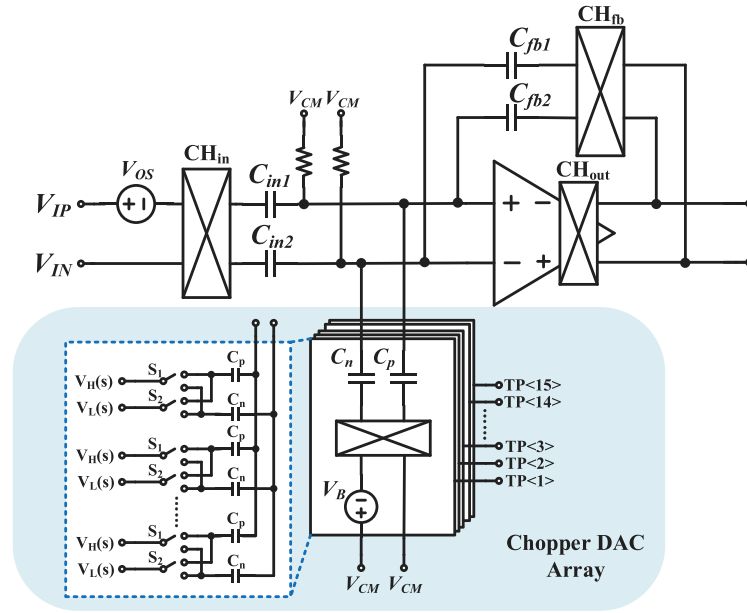


Fig. 5. Basic idea of the proposed SAC scheme.

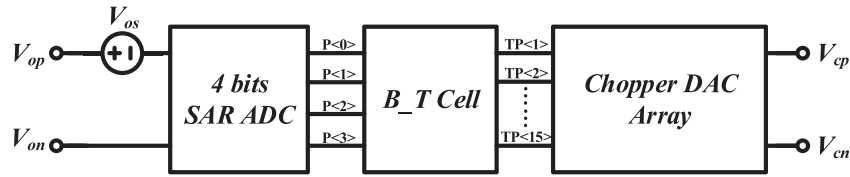


Fig. 6. Generation process of the differential compensation voltage.

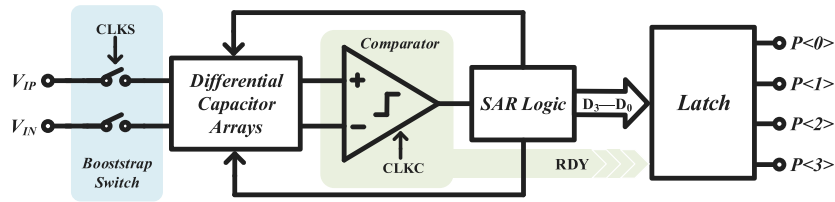


Fig. 7. Block diagram of the 4-bit SAR quantizer and latch.

voltage V_{cp} and V_{cn} . Firstly, the offset voltage V_{OS} at the output of the chopper amplifier is quantized by a 4-bit SAR quantizer to generate a corresponding 4-bit binary code. And then it is converted into a 15-bit thermometer code by B-T cell. The thermometer code is used to control the chopper DAC array for generating suitable compensation voltages, V_{cp} and V_{cn} .

4.1. SAR quantizer and latch array

The analog-to-digital converter is a bridge from analog to digital. In this design, the algorithm of successive approximation is adopted, and the analog signal at the chopper amplifier's output end is converted continuously into the corresponding 4-bit binary code by a 4-bit SAR quantizer.

Fig. 7 shows the block diagram of the 4-bit SAR quantizer and latch in this design. Considering the common-mode noise rejection capability and conversion accuracy, a differential structure is selected for this SAR quantizer. The SAR quantizer includes a bootstrap switch, a differential capacitor array, a dynamic comparator, and the SAR control logic. Its working process can be divided into two stages. The first stage is the sampling stage, and the capacitor array samples

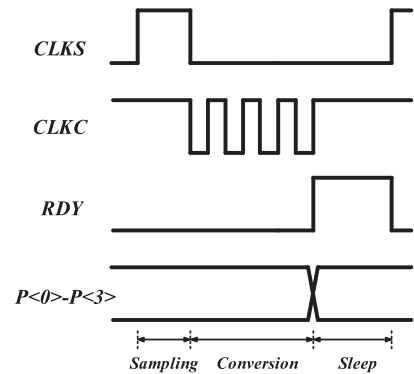


Fig. 8. Working timing diagram of the 4-bit SAR quantizer and latch.

the input signal through the bootstrap switch, and at the same time reduces signal distortion. The second stage is the conversion stage, and the comparator compares the outputs of the differential capacitance digital-to-analog converter (DAC) and passes the comparison result to

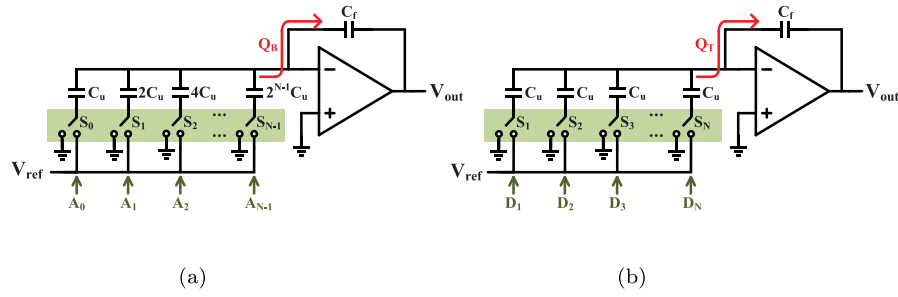


Fig. 9. Schematic diagram of capacitive DAC (a) In the case of binary codes (b) In the case of thermometer codes.

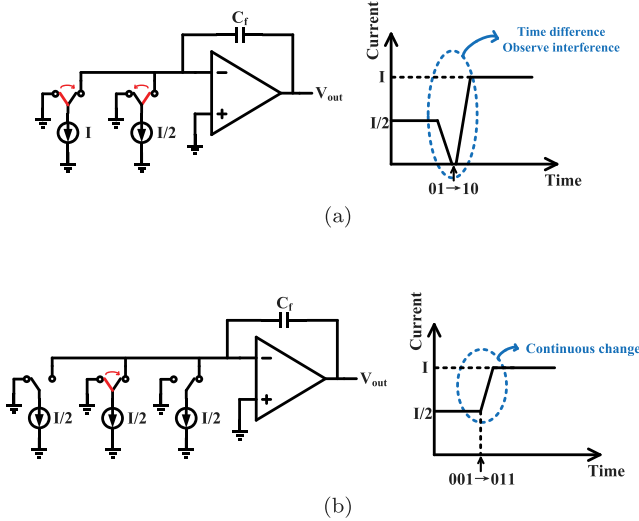


Fig. 10. Anti-interference ability comparison (a) binary-code DAC (b) thermometer-code DAC.

the SAR control logic unit. The SAR control logic unit obtains the corresponding 1-bit binary code according to the output of the previous stage comparator, and then the latch stores the binary code which will subsequently output. At the same time, the logic unit generates a control signal through the corresponding logic to control the output voltage of the differential capacitance DAC. Finally, the process of successive approximation is completed, and the input analog signal is quantized into a corresponding 4-bit binary code output. Its working timing diagram is shown in Fig. 8, where CLKS is the sampling clock signal, and CLKC is the comparator clock signal. When CLKS is pulled down, every time CLKC changes from up to down, the comparator performs a comparison. After four comparisons, CLKC pulls up, and at the same time, RDY also pulls up, indicating that the SAR quantizer has completed the quantization work and the generated binary signal can be latched by the latch array for subsequent application.

Although this work is not taped out, some discussion about the layout is included. To isolate the influence of the digital circuit noise on the analog circuit, analog and digital parts can retain a certain space. And the guarding ring can be added around the comparator which is more sensitive to the noise. The analog output value of the binary weighted charge redistribution D/A conversion circuit is directly related to the capacitance ratio. Therefore, special attention should be paid to the capacitor mismatch when designing the layout of the capacitor array in SAR ADC. In this design, the large capacitors can be composed of unit capacitors in parallel. At the same time, to reduce the parasitic effect of the capacitance routing, the method of local co-centroid can be used. And the dummy capacitors can be added to keep the environments of all capacitors consistent as far as possible.

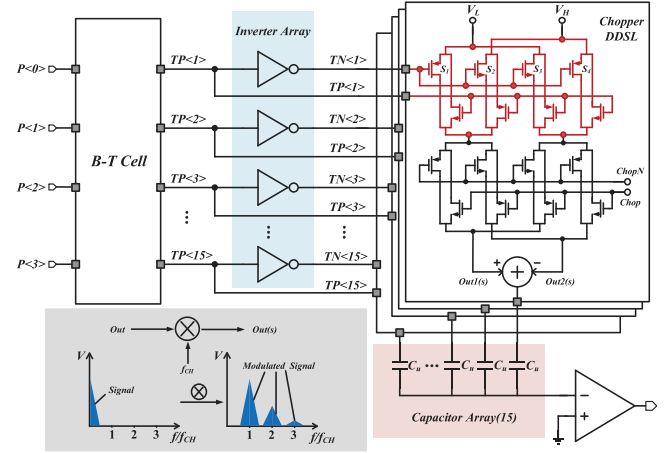


Fig. 11. Schematic diagram of the proposed new chopper DAC array.

4.2. B-T cell

After quantizing the output signal of the chopper amplifier by the SAR quantizer, a string of binary codes will be generated, but the compensation voltage that is used to eliminate the electrode DC offset is an analog quantity. So it is necessary to design a digital-to-analog converter (DAC) that takes data represented as digital codes and produces an equivalent analog output.

Fig. 9(a) shows a schematic diagram of a capacitive DAC controlled by binary codes. The binary digital signals \$A_0\$ to \$A_{N-1}\$ respectively control the switches \$S_0\$ to \$S_{N-1}\$. Each branch will store different charges due to different capacitance values, and finally, they can be combined to obtain different total charge values \$Q_B\$. Through the inverting operational amplifier and its \$C_f\$, \$Q_B\$ can be converted into an analog output voltage \$V_{outB}\$. In summary, the total charge value \$Q_B\$ is

$$Q_B = V_{ref} \sum_{i=0}^{N-1} 2^i A_i C_u \quad (3)$$

And then the output voltage \$V_{outB}\$ is

$$V_{outB} = -\frac{Q_B}{C_f} \quad (4)$$

Fig. 9(b) shows a schematic diagram of a capacitive DAC controlled by thermometer codes. The load capacitance of each branch is the same, so each branch will store the charges with the same weight. The thermometer digital codes \$D_1\$ to \$D_N\$ control the switches \$S_1\$ to \$S_N\$ respectively. In this way, the charges of each branch are controlled to store and release, and finally, different total voltage outputs will be obtained. In summary, the total charge value \$Q_T\$ is

$$Q_T = V_{ref} \sum_{i=1}^N D_i C_u \quad (5)$$

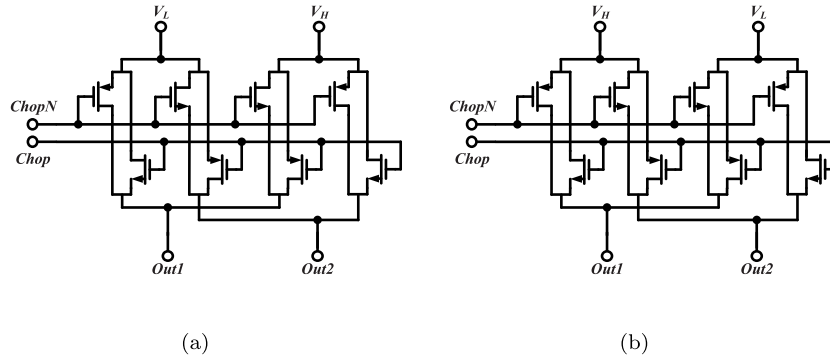


Fig. 12. Simplified diagram of Chopper DDSL when (a) TP=1 (b) TP=0.

And then the output voltage V_{outT} is

$$V_{outT} = -\frac{Q_T}{C_f} \quad (6)$$

In this design, the binary code is converted into the thermometer code, and the thermometer code is used to control the DAC to generate the corresponding analog output. The specific reasons are analyzed as follows. When using a DAC controlled by binary codes, a completely different voltage or current output at the moment of data switching will generate interference and noise in the analog signal output, as shown in Fig. 10(a). To reduce this interference, DAC controlled by thermometer codes is used. Thermometer codes use the amount of “1” to represent numbers, like counting on your fingers. DAC controlled by thermometer codes can continuously change when the data is switched, so it has an anti-interference ability, as shown in Fig. 10(b).

4.3. Chopper DAC array

In this design, a new chopper DAC array is proposed, as shown in Fig. 11. The 15-bit thermometer code $TP(1) - TP(15)$ generated by the B-T cell will generate the corresponding $TN(1) - TN(15)$ digital code through the inverter array, and each pair of thermometer codes (such as $TP(1)$ and $TN(1)$) will control a Chopper DDSL. The input voltage V_H and V_L will be modulated by the chopper circuit to generate the output voltage $Out1(s)$ and $Out2(s)$. And then $Out1(s)$ and $Out2(s)$ will be combined to generate the modulated compensation voltage. Finally, this compensation voltage will be fed back to the input end through the capacitor array to reduce the electrode offset voltage.

According to the thermometer codes being 0 or 1, Chopper DDSL in the chopper DAC array can be simplified into the following two chopper circuits. When $TP=1$ ($TN=0$), S_1 and S_4 are turned on, and the circuit can be simplified as shown in Fig. 12(a). When $TP=0$ ($TN=1$), S_2 and S_3 are turned on, and the circuit can be simplified as shown in Fig. 12(b).

5. Simulation results

A standard 0.18 μm CMOS process is used to verify the specified bio-signal analog front-end system. Over the required signal bandwidth, the CCIA is intended to produce a gain of 40 dB. The amplitude–frequency characteristic curves of this circuit at the TT, FF, and SS process corners are shown in Fig. 13. As can be observed, this circuit’s gain is roughly 39.15 dB, which is a little less than the amount intended due to the influence of parasitic resistances. The high-pass corner frequency is < 0.5 Hz at every PVT after using the SAC. DC offset from electrodes is therefore effectively minimized. Fig. 14 displays the transient simulation results of the comparator in the SAR quantizer, where V_{res} is the comparator input signal, which is the analog output of the capacitor DAC, and V_{outp} and V_{outn} are the comparator’s output signals. $CLKS$ is the sampling clock signal, and $CLKC$ is the comparator clock signal. The output node of the pre-amplification stage will be shorted to reset

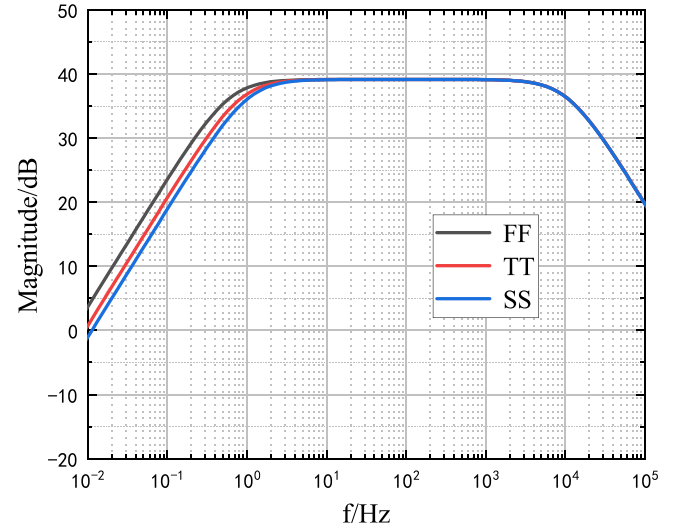


Fig. 13. System amplitude–frequency characteristics at different process corners.

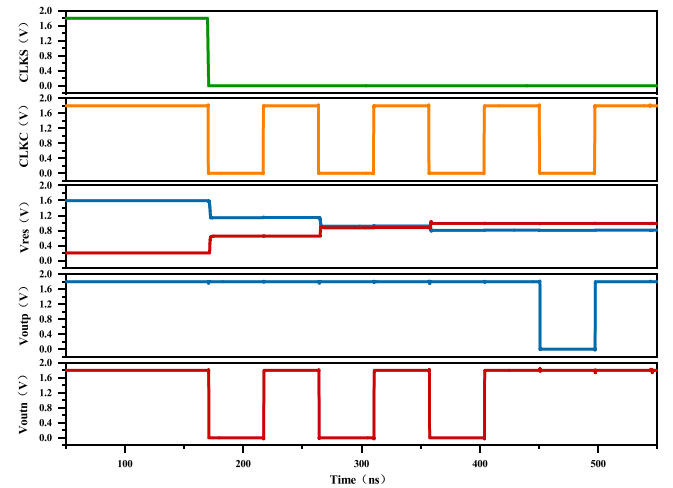


Fig. 14. Transient simulation waveform of the comparator.

when $CLKC$ is high, preventing the comparator’s memory effect. V_{outp} and V_{outn} are dragged up to VDD at the same moment. When $CLKC$ is low, the comparator compares the magnitude of the input voltages and using the latch’s positive feedback pulls one end of the comparator output up to VDD and the other end down to GND.

The input signal for the SAR quantizer’s dynamic test and analysis is a single-tone sine wave with a frequency of 508 Hz and an amplitude

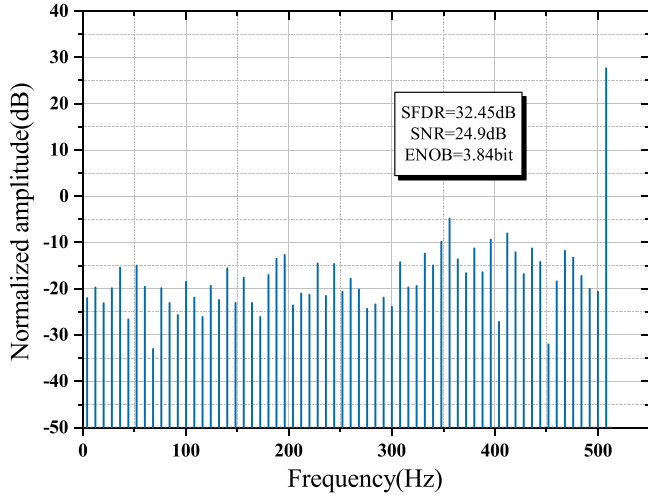


Fig. 15. Output FFT spectrum diagram of SAR quantizer.

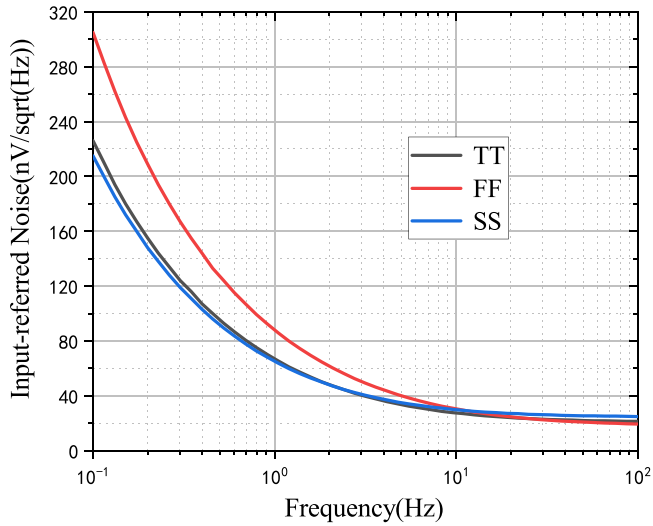


Fig. 16. Input-referred noise spectrum density within the bio-signal bandwidth.

of 1.5 V. The sampling frequency is set to 1.024 KHz. The resulting digital code of 256 points is then subjected to a quick Fourier analysis to obtain the FFT spectrum produced by the SAR quantizer, as illustrated in Fig. 15. The SAR quantizer's Effective Number of Bits (ENOB) is approximately 3.84 bits.

The CCIA input-referred noise spectrum density within the biosignal bandwidth is displayed in Fig. 16 using the chopper modulation technology. This leads to the conclusion that the integrated noise at the TT process corner within the bio-signal bandwidth (0.1–100 Hz) is 0.26 μVrms . The integrated noise at the FF process corner is 0.28 μVrms in the biosignal spectrum (0.1–100 Hz). The integrated noise at the SS process corner is 0.29 μVrms within the biosignal spectrum (0.1–100 Hz).

A transient simulation of the DC offset voltage at the output end is conducted to confirm the SAC circuit's suppression impact on the electrode DC offset, as shown in Fig. 17. When a 50 mV DC offset voltage is placed at the input end, the SAC circuit can compensate the DC offset voltage close to 0 in a relatively short amount of time. As illustrated in Fig. 18, a schematic diagram of the compensation procedure for the conventional analog DSL compensation technique is also provided. Comparing Figs. 17 and 18, it can be seen that the SAC circuit has a faster response time while also having a smaller chip area thanks to its straightforward circuit design and sparse use of capacitors.

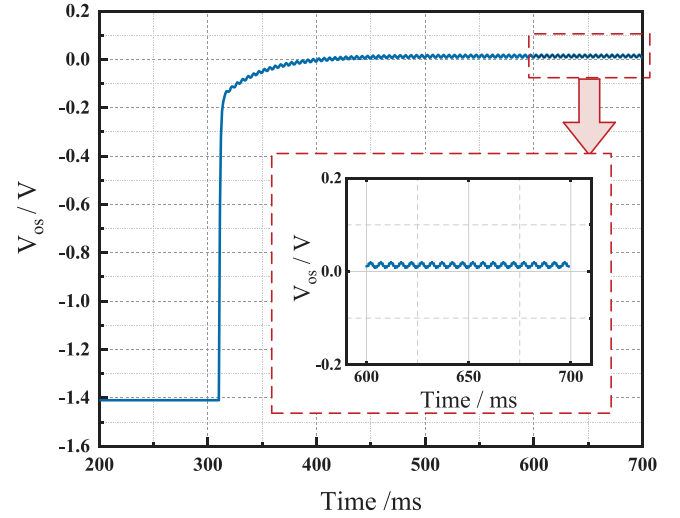


Fig. 17. Transient simulation diagram of the proposed SAC scheme.

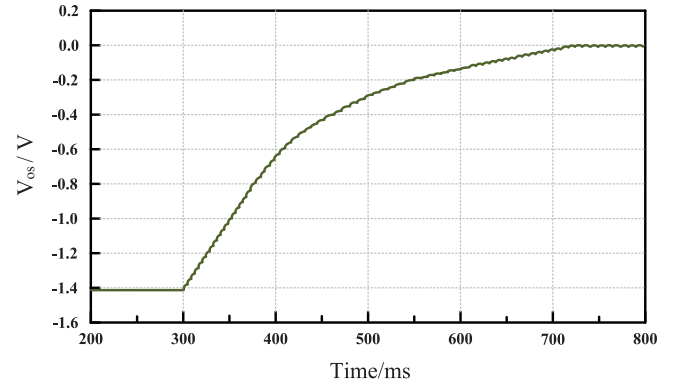


Fig. 18. Schematic diagram of the traditional analog DSL compensation scheme.

Noise Efficiency Factor (NEF) [16] is typically used in low frequency and low noise amplifier design to assess the relationship between noise, power consumption, and bandwidth:

$$NEF = V_{ni,rms} \sqrt{\frac{2I_{tot}}{\pi \cdot U_T \cdot 4KT \cdot BW}} \quad (7)$$

where $V_{ni,rms}$ is the equivalent input integrated noise of the circuit, BW is the noise integration bandwidth, I_{tot} is the current consumption of the whole system, and U_T is the thermal voltage KT/q . The NEF of the proposed circuit is calculated as 2.76.

The performance of the proposed analog front-end system for biopotential signals recording application is summarized in Table 1 and is compared with the state-of-art. This work demonstrates a great trade-off between noise and power consumption, as well as very low input-referred noise.

6. Conclusion

In this work, a SAC method is proposed to reduce the sampling electrode's DC offset voltage in the analog front-end circuit. This circuit prevents output saturation while lowering the output offset voltage of the chopper amplifier. It has been designed by the 0.18 μm CMOS process. 13.72 μW is used by the entire circuit from the 1.8 V power supply. At the TT process corner, the comparable integrated noise over a bandwidth of 0.1–100 Hz is 0.26 μVrms , while the NEF is 2.76. The DC offset caused by the non-ideal factors of the input chopper

Table 1
Performance summary and comparison with prior work.

Parameters	[17]	[18]	[19]	[20]	[21]	[22] ^a	This Work ^a
Process	0.13 μm	0.6 μm	0.18 μm	0.18 μm	0.18 μm	0.18 μm	0.18 μm
Supply Voltage	0.6 V	2.8 V	1.5 V	1.8 V	1.2 V	1.8 V	1.8 V
Gain	42 dB	39.4 dB	37 dB	40.04 dB	39.8–84.5 dB	34–52 dB	39.15 dB
Signal bandwidth	10–7 kHz	0.36–1.3 kHz	0.8–300 Hz	0.05–11 kHz	0.35–150 Hz	0.5–1 kHz	0.5–10 kHz
Input-referred noise	3.976 μVrms (10–7 kHz)	3.07 μVrms (0.5–1.3 kHz)	11.8 μVrms (1–1 kHz)	0.78 μVrms (0.05–11 kHz)	1.12 μVrms (0.35–150 Hz)	1.36 μVrms (0.5–100 Hz)	0.26 μVrms (0.1–100 Hz)
Power	2.4 μW	2.4 μW	1.8 μW	78.84 μW	1.44 μW	22.5 μW	13.72 μW
Electrode DC offset Tolerance	N.I.	N.I.	N.A.	N.I.	N.I.	50 mV	50 mV
NEF	3.66	3.09	16.5	1.88	2.7	18.5	2.76

N.I. stands for not included.

N.A. stands for not available.

^aSimulation Results.

is controlled without changing the main chopper amplifier's quiescent operating point. In addition, there are not many capacitors and the response time is quick, which will significantly lower the area consumption. This instrumentation amplifier is well suited for use in systems that acquire biological signals as well as other related applications.

CRedit authorship contribution statement

Tingyu Wang: Methodology, Investigation, Writing – original draft & editing. **Kui Wen:** Conceptualization, Writing – review. **Longjie Zhong:** Data curation, Formal analysis. **Shubin Liu:** Supervision, Visualization.

Declaration of competing interest

We wish to confirm the following facts that there are no known conflicts of interest associated with this manuscript, titled as “A 0.26 μVrms Instrumentation Amplifier Based on Discrete-Time DC Servo Loop with Chopper DAC Array”, and there has been no significant financial and personal relationships with other people or organizations that can inappropriately influence the outcome of this work.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work, and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. By doing this, we confirm that we have followed the regulations of our institutions concerning intellectual property.

Data availability

Data will be made available on request.

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